

Q How many weeks does it take for my baby's face to form?

By the time you go for your ultrasound at around 12 weeks, all your baby's facial features will be clearly recognizable—her face has been developing since the early weeks after conception.

BABY'S FACIAL FEATURE DEVELOPMENT UP TO 12 WEEKS

The following table shows some of the major milestones in the development of her facial features up to the end of the first trimester:

WEEK	FEATURES
6	» Your embryo develops dark spots on each side of her head that will become her eyes. Firstly they form "optic cups" that link to her brain via an optic stalk. The cups eventually become her retinas (the "screens" at the backs of her eyes) and irises (the colored parts of each eye). » Minuscule indents on either side of her head are the beginnings of your baby's ears. Inner tubes that will transmit sound in your baby's ears begin to form.
7	» Her eyes start to form lenses; these are created when certain cells of the ectoderm layer fold over. Eyelids begin to form from folds of skin either side of each eye. » Skin at the sides of your baby's head begin to fold over, so that her ears begin to form. » Nostrils appear at the front of the face, and her nose starts to take shape.
8	» Changes in the ectoderm layer of your embryo mean that by eight weeks the two sides of her face have come together to shape her mouth. » Pigmentation (which gives your baby's eyes their color) starts to collect in the iris of her eyes (see p.168). » Your baby's tongue starts to form, initially above the plate that will become the roof of her mouth.
9	» Your baby's eyes can now move around in their sockets. » Her tongue has developed taste buds.
10	» Your baby's nose starts to protrude, giving her face a more human profile. » Her eyelids start to close over her eyeballs, protecting them from light coming through your abdominal wall. This light could damage the developing retinas. » Your baby's mouth and lips are now fully formed. » Your baby's outer ears now look exactly as they will at birth, but there's still some work to be done inside before she can actually hear.
11	» The roof of your baby's mouth—her palate—is complete. » Tooth buds begin to appear inside her mouth. These will become her milk and adult teeth when the two rows eventually separate.
12	» Your baby's eyes and ears have moved upward as her head has grown, and her eyes have also moved toward the front of her face and are now more or less in their final positions. Your baby can use her mouth, jaw, and tongue to swallow and yawn.

Q I've been singing to my baby, but can she actually hear me? And can she hear my partner singing, too?

Most doctors agree that by the time a fetus is 14 weeks old, she responds to familiar sounds with a quickening of her heart rate. Certainly by 16 weeks, the three bones that transmit sound in her inner ear and the auditory pathways that take sound to her brain are developed enough to suggest that she can hear what's going on inside your body. So she'll hear the swooshing of your blood, your heartbeat, and your tummy rumbles. By 24 weeks, studies show that she'll turn her head in response to a familiar sound—in particular the sound of your voice. Sounds outside the uterus are quieter and more muffled, so it's harder for her to make them out. But hearing a voice other than yours, even if it sounds as if it's being transmitted through water, is enough to make her familiar with it, so your partner's singing definitely isn't in vain.

Q If my baby is developing taste buds, does that mean she can taste the amniotic fluid?

Absolutely! At around the same time that she learns to swallow (roughly 16 weeks pregnant), she develops receptors on her tongue that mean her taste buds can start interpreting the flavors of the amniotic fluid. Contrary to what you might think, your amniotic fluid isn't neutral tasting: it's salty and is infused with flavors from whatever you've been eating, particularly strong flavors such as spices, onion, and garlic. Think of this as a good opportunity to get your baby used to some of your favorite foods—and to the healthy ones that you'll want her to eat when she's older!

Q When does my baby get her skin and what does it look like?

Skin begins to form from the moment a blastocyst becomes an embryo (see p.152), but until well into the second trimester, it's thin and translucent so that nutrients from the amniotic fluid can easily pass through its cells. By 15 weeks pregnant, this translucent covering is